LINUX INTERVIEW - 50 MOST ASKED QUESTIONS

PAGE 1
Q1 - Q10

BASIC CONCEPTS & FUNDAMENTALS

1. What is Linux?

Ans: Linux is an open-source operating system kernel created by Linus Torvalds. It manages hardware, memory, processes, filesystems and networking, and provides services to applications through system calls.

Interview Tip: Say it as: "Linux is the kernel that acts as a bridge between hardware and software."

2. What is the Linux Kernel?

Ans: The kernel is the core component of Linux. It interacts with hardware and manages system resources like CPU, memory, devices, filesystems, networking and processes. It also provides system calls interface to applications.

Key Point: Kernel = Heart of Linux.

3. What is the difference between Linux and Unix?

Ans:
- Unix is a family of operating systems that originated in the 1970s.
- Linux is an open-source, Unix-like kernel.
- Unix is mostly proprietary; Linux is open-source.
- Linux distributions include tools and utilities to make it a complete OS.

Best Answer: "Linux is Unix-like but not Unix."

4. What is a Linux Distribution?

Ans: A Linux Distribution is a complete OS that includes the Linux kernel along with system tools, libraries, package manager and applications.

Examples: Ubuntu, Debian, RHEL, CentOS Stream, Rocky Linux, AlmaLinux, Fedora, openSUSE.

5. What is the difference between a Shell and a Terminal?

Ans:
- Terminal: It is the interface (window) through which we interact with the system.
- Shell: It is the program that interprets our commands and provides the interface.

Example Shells: bash, zsh, sh, ksh, fish.

6. Who is the Root User?

Ans: Root is the superuser in Linux with the highest privileges. It can modify or access any part of the system.

Best Practice: Use 'sudo' for admin tasks instead of logging in as root.

7. What is the Linux Filesystem Hierarchy (FHS)?

Ans: Linux organizes files in a tree-like structure. Important directories:

/	- Root directory
/bin	- Essential user binaries
/sbin	- System binaries
/etc	- Configuration files
/home	- User home directories
/var	- Variable files (logs, spool, cache)
/tmp	- Temporary files
/usr	- User programs and data
/proc	- Process and kernel info (virtual)
/dev	- Device files
/root	- Root user's home directory
/opt	- Optional third-party software

8. What is /proc?

Ans: /proc is a virtual filesystem that provides information about running processes and the system. Data is not stored on disk.

Example: cat /proc/cpuinfo , cat /proc/meminfo

9. What is an Inode?

Ans: Inode stores metadata of a file (owner, permissions, size, timestamps, pointers to data blocks, etc.). Filename is stored in directory which points to inode.

Command: ls -i filename

10. What is the difference between Hard Link and Symbolic Link?

Ans:
- Hard Link: Points to the same inode. Multiple filenames for the same file. Cannot cross filesystems and cannot link directories.
  Command: In original hardlink
- Symbolic Link (Soft Link): Points to the path of another file. Can cross filesystems and can link directories.
  Command: ln -s original symlink

☆ INTERVIEW TIP: Always give answers in your own words, with simple examples and real-life use. Interviewers value clarity and practical understanding more than memorized definitions.

Next Page (Page 2) →
- Users & Permissions
- Processes
- Systemd & Services

LINUX INTERVIEW - 50 MOST ASKED QUESTIONS

11. What does 'chmod 755' mean?

Ans: It sets permissions as:

Owner (7)	Group (5)	Others (5)
rwx (read, write, execute)	r-x (read, execute)	r-x (read, execute)

Command: `chmod 755 file.sh`

Interview Answer: It gives full permission to owner and read + execute permission to group and others.

12. What is the difference between chmod, chown and chgrp?

Ans:
- chmod → changes permissions.
- chown → changes owner (and group).
- chgrp → changes group.

Example Commands:

```
chmod 644 file.txt              # change permissions
chown user1:group1 file.txt     # change owner and group
chgrp group1 file.txt           # change group only
```

13. What is the difference between read, write and execute permission for a file and a directory?

Ans: For a file:
- r (read) → read the content
- w (write) → modify the content
- x (execute) → run the file as a program/script

For a directory:
- r (read) → list files in the directory
- w (write) → create, delete, rename files in it
- x (execute) → access / enter the directory

14. What is umask?

Ans: umask defines the default permission mask that is removed from newly created files and directories.

Check: `umask`

Example: Default file permission = 666
If umask = 022 → Final permission = 644

15. What is the sticky bit?

Ans: Sticky bit on a directory restricts users from deleting or renaming files they don't own, even if they have write permission.

Check: `ls -ld /tmp`

Example: `drwxrwxrwt` (t at the end indicates sticky bit)

Interview Answer: Used on world-writable directories like /tmp to prevent accidental or malicious deletion.

16. What is sudo?

Ans: sudo allows authorized users to execute commands with superuser (root) privileges.

Command: `sudo command`

Example: `sudo systemctl restart nginx`

Interview Answer: It provides a secure way to perform admin tasks without logging in as root.

17. How do you create a new user?

Ans:

Command:
```
sudo useradd -m username
sudo passwd username
```

Explanation: -m creates home directory
passwd sets the user password

18. How do you add a user to a group?

Ans:

Command: `sudo usermod -aG groupname username`

Explanation: -a → append the group without removing existing groups.

19. Where is user information stored in Linux?

Ans:
- /etc/passwd → user account information
- /etc/shadow → encrypted passwords
- /etc/group → group information
- /etc/gshadow → group passwords

20. How do you find which groups a user belongs to?

Ans:

Command:
```
id username
   or
groups username
```

Example Output:
```
uid=1001(user1) gid=1001(user1)
groups=1001(user1),27(sudo),1002(devs)
```

Interview Answer: This shows primary group (gid) and all supplementary groups.

INTERVIEW TIP

1. Explain with real-life examples.
2. Show that you understand "why" behind each command.
3. Speak clearly, be concise and structured.

COMING NEXT (Page 3)
- Users & Groups
- Processes
- Systemd & Services
- Logs

Keep Practicing. Keep Learning. You've Got This! ☺

LINUX INTERVIEW - 50 MOST ASKED QUESTIONS

PROCESSES, JOBS & SCHEDULING

21. What is a process in Linux ?

Ans: A process is a running instance of a program with its own memory space, resources and process ID (PID).

Command : `ps aux` # view running processes

Interview Answer: A process is an active program in memory with its own PID, resources and execution context.

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22. What is PID 1 ?

Ans: PID 1 is the first user space process started by the kernel during boot. In modern Linux it is 'systemd'.

Command : `ps -p 1 -o pid,comm,args`

Interview Answer: PID 1 is the parent of all processes and responsible for system initialization and service management.

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23. What is a zombie process ?

Ans: A zombie process is a terminated process that still has an entry in the process table because its parent has not collected its exit status.

Check : `ps aux | grep Z`

Interview Answer: A zombie process (Z state) is a process that has finished execution but remains in the process table until its parent reads the exit status.

---

24. What is the difference between foreground and background process ?

Ans: Foreground : Runs in the current terminal and takes input/output.

Background : Runs in the background without blocking the terminal.

Commands :

```
Run in background  : command &
Move to background : Ctrl + Z  then  bg
Bring to foreground : fg
```

Interview Answer: Foreground process occupies the terminal, while background process runs without blocking it.

---

25. What is the difference between SIGTERM and SIGKILL ?

Ans: SIGTERM (15) : Graceful termination. Gives the process time to clean up. Can be caught and handled.

SIGKILL (9) : Forcefully kills the process immediately. Cannot be caught or ignored.

Commands :

```
kill -15 <PID>    # SIGTERM
kill -9  <PID>    # SIGKILL
```

Interview Answer: I always try SIGTERM first for safe shutdown, and use SIGKILL only when the process doesn't stop.

---

26. How do you check CPU and memory usage ?

Ans: Use system monitoring tools to check resource usage.

Commands :

```
top / htop                      # real-time process view
free -h                         # memory usage
ps aux --sort=-%cpu | head      # top CPU processes
ps aux --sort=-%mem | head      # top memory processes
vmstat 1 5                      # system statistics
```

Interview Answer: I use top/htop for real-time view and free, ps, vmstat for detailed CPU and memory analysis.

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27. What is load average ?

Ans: Load average shows the average number of runnable or uninterruptible tasks in the last 1, 5 and 15 minutes.

Command : `uptime`

Interview Answer: Load average indicates system workload. It is not CPU usage; it shows how many processes are waiting for CPU or I/O.

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28. How do you find a process using a specific port ?

Ans: Use network tools to identify the process listening on a port.

Command :

```
ss -tulnp | grep :80      # check port 80
lsof -i :80               # alternative way
```

Interview Answer: I use ss (or lsof) with the port number to find the PID and process name using that port.

---

29. How do you change the priority of a process ?

Ans: Use nice and renice to adjust process priority.

Commands :

```
nice -n 10 command        # start with lower priority
renice -n 5 -p <PID>      # change running process priority
```

Interview Answer: I use nice to start a process with a specific priority and renice tochange the priority of an already running process. Lower value means higher priority.

---

30. What is a cron job ? How do you manage it ?

Ans: Cron is a time-based job scheduler used to run commands or scripts automatically at scheduled intervals.

Commands :

```
crontab -e                  # edit user cron jobs
crontab -l                  # list user cron jobs
systemctl status crond      # check cron service (RHEL/CentOS)
systemctl status cron       # check cron service (Debian/Ubuntu)
```

Interview Answer: Cron is a scheduler used to automate tasks like backups and monitoring. I manage it using crontab for user-level jobs and systemctl for the cron service.



INTERVIEW TIP :

- Don't just give commands - explain what is happening and why.
- Use real-life examples from your practice or projects.
- Speak clearly and keep the answer simple but technically correct.

Next Page (Page 4) →
- File Search & Text Processing
- Disk Usage & Storage
- Package Management

LINUX INTERVIEW - 50 MOST ASKED QUESTIONS

31. How do you check IP address of a Linux system?

Ans: You can use ip command to view IP address.

Commands:
```
ip addr
or
ip -br addr          # brief format
```

Interview Answer:
I use 'ip addr' to view all network interfaces and their IP addresses.

32. How do you check listening ports on a system?

Ans: Use ss command to see listening ports and the services using them.

Command:
```
ss -tulnp            # shows TCP, UDP, listening, ports, process
```

Example:
```
ss -tulnp | grep :80     # check port 80
```

Interview Answer:
I use ss -tulnp to check which ports are listening and which process is using them.

33. How do you test network connectivity?

Ans: Use different tools depending on what you want to test.

```
ping google.com              # check basic connectivit
curl http://example.com      # check HTTP/website
nslookup example.com         # check DNS resolution
traceroute example.com       # check network path
ping -c 4 8.8.8.8            # test external connectivity
```

Interview Answer:
I first check basic connectivity using ping, then verify the service using curl, and if needed I check DNS and route.

34. What is DNS?

Ans: DNS (Domain Name System) converts human readable domain names into IP addresses.

Common DNS Records:

A	→	Maps domain to IPv4 address
AAAA	→	Maps domain to IPv6 address
CNAME	→	Alias record
MX	→	Mail server record
TXT	→	Text information (SPF, verification, etc.)

Interview Answer:
DNS is like the phonebook of the internet. It helps systems find the IP address of a domain name.

35. How do you check the default gateway and routing table?

Ans: Use ip route command.

Commands:
```
ip route                 # show routing table
ip route show default    # show default gateway
```

Interview Answer: I use ip route to check the routing table and find the default gateway.

36. What is the difference between ifconfig and ip command?

Ans: ifconfig is an older tool (from net-tools package). ip command is newer, more powerful and preferred.

ifconfig (Old)	ip (New)
Part of net-tools package	Part of iproute2 package
Limited features	More features and flexibility
Deprecated in many distros	Actively maintained
Example: ifconfig eth0	Example: ip addr show dev eth0

Interview Answer: I use ip command because it is modern, powerful and works with all network configurations.

37. What is the difference between df and du?

Ans:
- df shows disk space usage of file systems.
- du shows disk usage of files and directories.

Commands:
```
df -h                # filesystem usage
du -sh /path         # directory size
```

Interview Answer:
If I want to see free space on disk, I use df. If I want to see how much space a directory is using, I use du.

38. Why df shows 100% full but du shows less usage?

Ans: It usually happens when a file is deleted but still open by a running process. The space is not released until the process closes it.

Check:
```
lsof +L1         # list open files that are deleted
```

Interview Answer: I check with lsof +L1 and restart the process or service to free the space.

39. What is LVM?

Ans: LVM (Logical Volume Manager) provides flexibility in managing disk storage. It allows resizing, snapshots and easy management.

LVM Structure:
Physical Volume (PV) → Volume Group (VG) → Logical Volume (LV) → File System

Common Commands:
```
pvdisplay        # show physical volumes
vgdisplay        # show volume groups
lvdisplay        # show logical volumes
lvextend         # extend logical volume
resize2fs        # resize filesystem
```

Interview Answer: I use LVM to manage storage flexibly without downtime or data loss.

40. How do you check mounted file systems?

Ans: Use the following commands.

Commands:
```
df -hT           # show all mounted filesystems
findmnt          # shows mount tree in a nice format
mount            # shows currently mounted filesystems
```

Interview Answer: I use df -hT for quick view and findmnt to understand the mount hierarchy.

INTERVIEW TIPS ☆

- ☑ Always explain before giving command.
- ☑ Start with identify → analyze → resolve approach.
- ☑ Think out loud. Interviewers value your thought process.
- ☑ Practice common commands regularly.

IMPORTANT COMMANDS SUMMARY

```
Network  : ip, ss, ping, curl, traceroute
Disk     : df, du, findmnt, lsof
Process  : ps, top, htop, kill
System   : uptime, free, vmstat, dmesg
Service  : systemctl, journalctl
```

COMING NEXT (Page 5) →

- Users & Groups
- Permissions Deep Dive
- File Types
- Links and Inodes
- More Interview Questions

☆ Learn. Practice. Troubleshoot. Repeat. ☆

LINUX INTERVIEW - 50 MOST ASKED QUESTIONS

41. How do you find a file in Linux ?

Ans: Use find command.

Commands :

```
find / -name file.txt          # find by name
find / -type f -name "*.log"   # find log files
find /home -mtime -7           # modified in last 7 days
find / -size +100M             # files larger than 100MB
find / -user username          # files owned by a user
```

Interview Answer :

I use find command with different options like name, type, size, user and time to locate files.

42. How do you find text inside files ?

Ans: Use grep command.

Commands :

```
grep "error" /var/log/app.log     # in a file
grep -R "error" /var/log/         # recursive search
grep -i "error" *.log             # case insensitive
grep -n "error" app.log           # with line number
```

Interview Answer :

I use grep to search text. -R for recursive, -i for case-insensitive and -n to show line numbers.

43. How do you check disk usage and inode usage ?

Ans: df command shows disk space usage.

- df -i shows inode usage.
- du shows directory/file size usage.

Commands :

```
df -h              # disk space
df -i              # inode usage
du -sh /var/*      # size of directories
du -h /path/file   # size of a file
```

Interview Answer :

I use df for disk space, df -i for inode usage and du to find which directory is consuming space.

44. What is a log file and why is it important ?

Ans: A log file records events, errors, warnings and activities of system or applications.
It helps in troubleshooting and monitoring.

Common Log Locations :

```
/var/log/messages      # system messages
/var/log/syslog        # general system logs
/var/log/auth.log      # authentication logs (Debian/Ubuntu)
/var/log/secure        # authentication logs (RHEL/CentOS)
/var/log/dmesg         # kernel ring buffer logs
/var/log/cron          # cron job logs
```

Interview Answer :

Logs help me understand what happened on the system and why an issue occurred. They are the first place I check during troubleshooting.

INTERVIEW TIPS ☆

- ☑ Understand the concept
- ☑ Know the command
- ☑ Know real-world use
- ☑ Explain clearly and confidently
- ☑ Practice scenarios

Quick Command Cheat Sheet

File Search	:	find
Text Search	:	grep
Process	:	ps, top, htop
Disk	:	df, du, lsblk
Network	:	ip, ss, ping
Service	:	systemctl
Logs	:	journalctl, tail

45. How do you kill a process in Linux ?

Ans: Use kill command.

Commands :

```
kill <PID>              # send SIGTERM (15)
kill -9 <PID>           # send SIGKILL (9)
pkill process_name      # kill by process name
killall process_name    # kill all processes by name
```

Interview Answer :

I first try SIGTERM (kill) to allow graceful shutdown. If it doesn't stop, I use SIGKILL (-9).

46. How do you set a password policy ?

Ans: Password policies are set in /etc/login.defs or using PAM modules.

Key File: /etc/login.defs

Important Parameters :

```
PASS_MAX_DAYS      # maximum days before password change
PASS_MIN_DAYS      # minimum days between changes
PASS_MIN_LEN       # minimum password length
PASS_WARN_AGE      # warning before password expires
```

Interview Answer :

I set password rules in /etc/login.defs or with PAM to enforce strong password policies.

47. What is cron and how does it work ?

Ans: Cron is a time-based job scheduler in Linux. It runs commands or scripts automatically.

Commands :

```
crontab -e          # edit current user crontab
crontab -l          # list current user crontab
crontab -r          # remove current user crontab
ls /etc/cron.*      # system-wide cron jobs
```

Common Cron Format :

```
* * * * *  command_to_execute
| | | | |
| | | | day (0-6, Sun-Sat)
| | | month (1-12)
| | day (1-31)
| hour (0-23)
min (0-59)
```

Interview Answer :

Cron helps automate repetitive tasks. I use crontab for user jobs and /etc/cron.d or cron.* for system jobs.

48. How do you backup files in Linux ?

Ans: Use tar, rsync or cp commands based on need.

Commands :

```
tar -czvf backup.tar.gz /path/to/dir   # backup with tar
rsync -av /source/ /backup/            # incremental backup
cp -r /source/ /backup/                # simple copy backup
```

Interview Answer :

I use tar for compressed backup, rsync for incremental backup and cp for simple copy.

49. How do you schedule a one-time task ?

Ans: Use at command.

Commands :

```
echo "command" | at 10:00 PM tomorrow
atq                # list scheduled jobs
atrm <job_id>      # remove scheduled job
```

Interview Answer :

at is used to schedule a one-time task at a specific time.

50. What would you do if you forget root password ?

Ans: Steps (General):

1. Reboot the system.
2. At GRUB menu, select recovery mode or single user mode.
3. Remount filesystem in read-write mode.
4. Run passwd and set new root password.
5. Reboot the system.

Interview Answer :

I boot into recovery mode, reset the root password using passwd command, then reboot.

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